

Don't Repeat Yourself!

how custom Python modules in Microsoft Fabric gives you back hours every day



MICROSOFT

FABRIC

A scenic landscape featuring a calm lake reflecting the surrounding mountains and sky. In the foreground, a wooden house with a shingled roof sits on stilts over the water. The background shows steep, rocky mountains with patches of greenery and some clouds. The overall mood is peaceful and majestic.

Bas Land

Don't Repeat
Yourself

Today

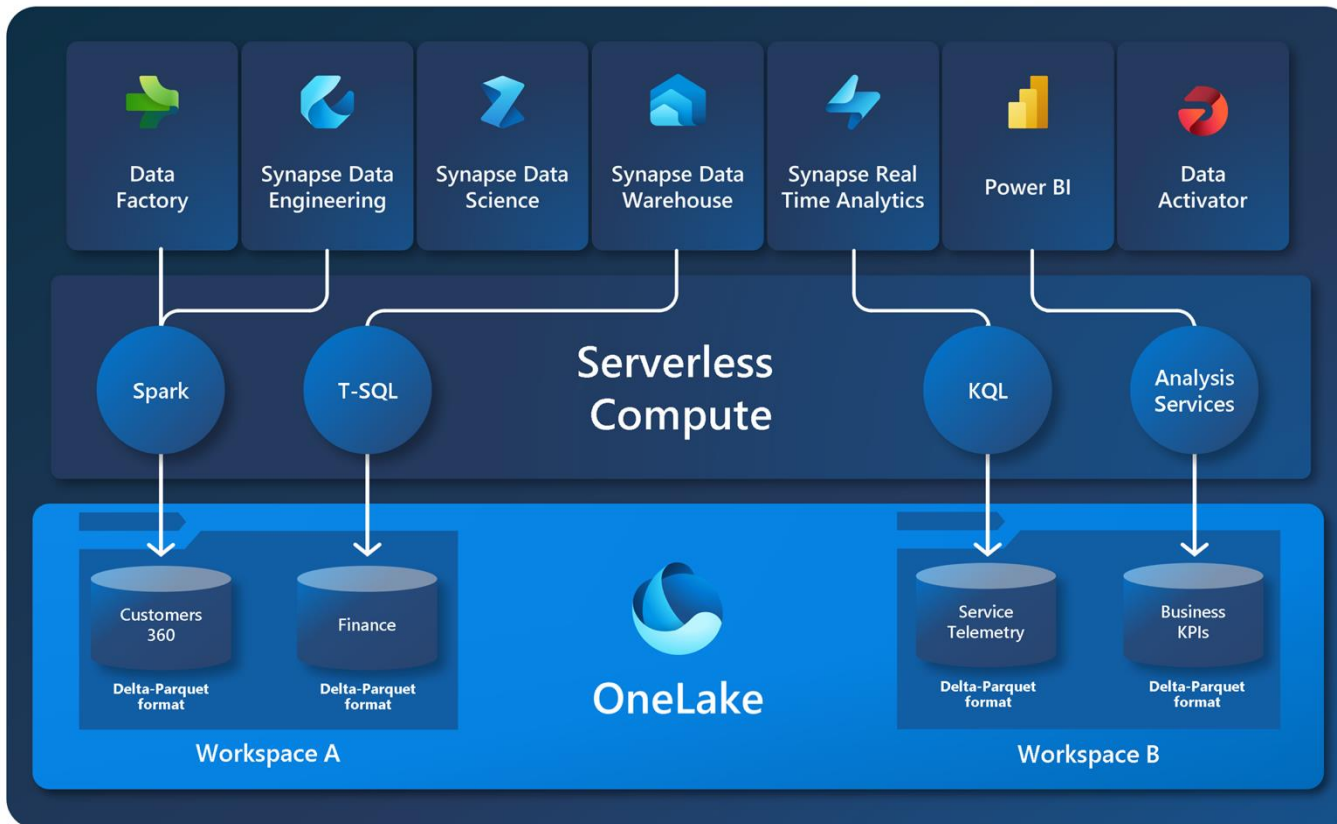
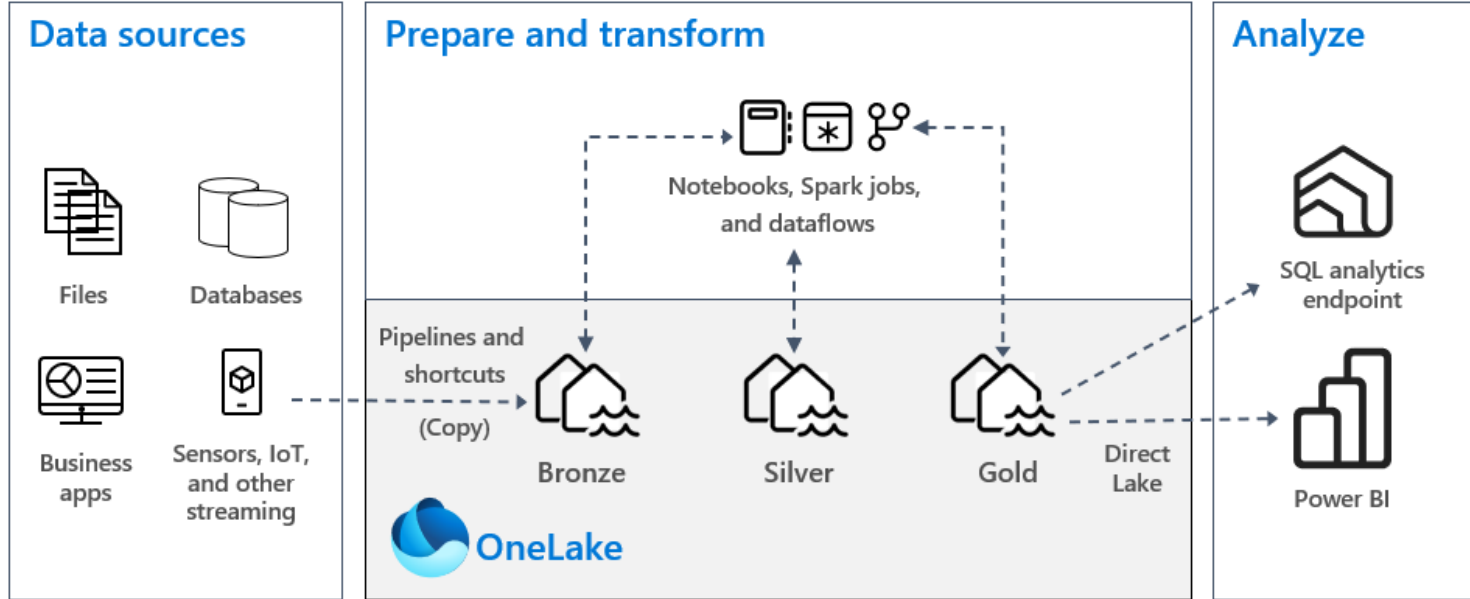


- Core concepts of Fabric compute
- The Problem: Repetitive Coding
- The Solution: Reusable Python Modules
- Live Demo
- Q&A



Core concepts
Fabric Compute

Fabric compute



Python in Fabric



- Python and Spark work together quite nicely!
- Benefits of using Python:
 - Extensive libraries for common tasks in data engineering, data science, statistics
 - Community support (probably the most popular programming language)
 - Flexibility in data engineering tasks

Python examples (1)



```
1 one = 1
2 two = 2
3 total = one + two
4 message = f"Hello world. The total is {total}."
5
6 print(message)
```



```
1 one = 1
2 two = 2
3 total = one + two
4 message = f"Hello world. The total is {total}."
5
6 print(message)
```

[129] ✓ 1 sec - Command executed in 302 ms by Olav van Rabenswaaij | Kimura on 3:08:32 PM, 11/18/24

... Hello world. The total is 3.

Python examples (2)



```
1 columns = ['ID', 'Name', 'Age']
2 data = [
3     (1, 'Alice', 28),
4     (2, 'Bob', 35),
5     (3, 'Charlie', 23)
6 ]
7
8 df = spark.createDataFrame(data, columns)
9
10 df.show()
```

[2] ✓ 2 sec - Command executed in 1 sec 595 ms by admin-kimura on 4:29:33 PM, 11/18/24

> Spark jobs (3 of 3 succeeded) Resources Log

```
... +---+-----+---+
    | ID|   Name|Age|
    +---+-----+---+
    |  1|  Alice| 28|
    |  2|   Bob| 35|
    |  3|Charlie| 23|
    +---+-----+---+
```

Python examples (3)



```
1 import re
2 from datetime import datetime
3
4 #Table to load
5 tables = [
6     {'endpoint': 'invoices', 'table_key': "['id']", 'json_root_array': 'results'},
7     #{'endpoint': 'tasks', 'table_key': "['id']", 'json_root_array': 'results'},
8     {'endpoint': 'registrations', 'table_key': "['id']", 'json_root_array': 'results'},
9     {'endpoint': 'registrations/pricelistitems', 'table_key': "['id']", 'json_root_array': ''},
10    {'endpoint': 'relations', 'table_key': "['uniqueIdentifier']", 'json_root_array': 'results'},
11    {'endpoint': 'users', 'table_key': "['id']", 'json_root_array': 'results'},
12    #{'endpoint': 'users/teams', 'table_key': "['id']", 'json_root_array': ''},
13    {'endpoint': 'companies', 'table_key': "['id']", 'json_root_array': ''}
14 ]
15
16 #Work
17 activities = []
18
19 #Load data to Silver layer delta tables
20 for table in tables:
21     endpoint = table['endpoint']
22     table_key = table['table_key']
23
24     args = {
25         'target_table': endpoint,
26         'source_system': "AdminPulse",
27         'file_type': "json",
28         'table_key_list': table_key,
29         'timestamp': timestamp
30     }
31
32     activity = {
33         'name': endpoint,
34         'path': 'AP_Silver_Worker_FullLoad',
35         'args': args,
36         'timeoutPerCellInSeconds': 9000,
37         'retry': 1,
38         'retryIntervalInSeconds': 10,
39         'dependencies': []
40     }
41     activities.append(activity)
```



The problem:

Repetitive coding

Repetitive code



- Common challenges:
 - Duplicated code across processes
 - Manual changes leading to errors
 - Difficulty in scaling
- Impact on productivity and developer's happiness



Real-world scenario

- Extending a dimension table in the data warehouse
- Extracting a couple of new tables
 - Making extract scripts
 - Making scripts for the Silver/persistent layer
- Adding this data to the dimension
 - Adding the join
 - Putting new fields in the mapping
 - Also in the insert/update merge, in the unknown member, etc
- Lots of work!



Cost of repetition

- Time spent on writing and debugging repetitive code
- Increased risk of inconsistencies and bugs
- Difficulty maintaining and extending the codebase

Don't repeat yourself (DRY)



- A principal aimed at reducing repetition
- Important in software development – let's learn from this as data engineers
- Enhances code maintainability
- Facilitates scalability

Don't repeat yourself (DRY)



- Every piece of knowledge must have a **single**, **unambiguous**, **authoritative** representation within a system
- Single: everything should be configured in just one location/script
 - Examples: connection strings, file paths, etc
- Unambiguous: everything should be clearly named and documented
 - Examples: Python docstrings
- Authoritative: single source of truth for data models, objects, etc
 - Examples: Process/notebook to generate Silver layer tables



The solution:
Reusable Python modules



Custom Python modules

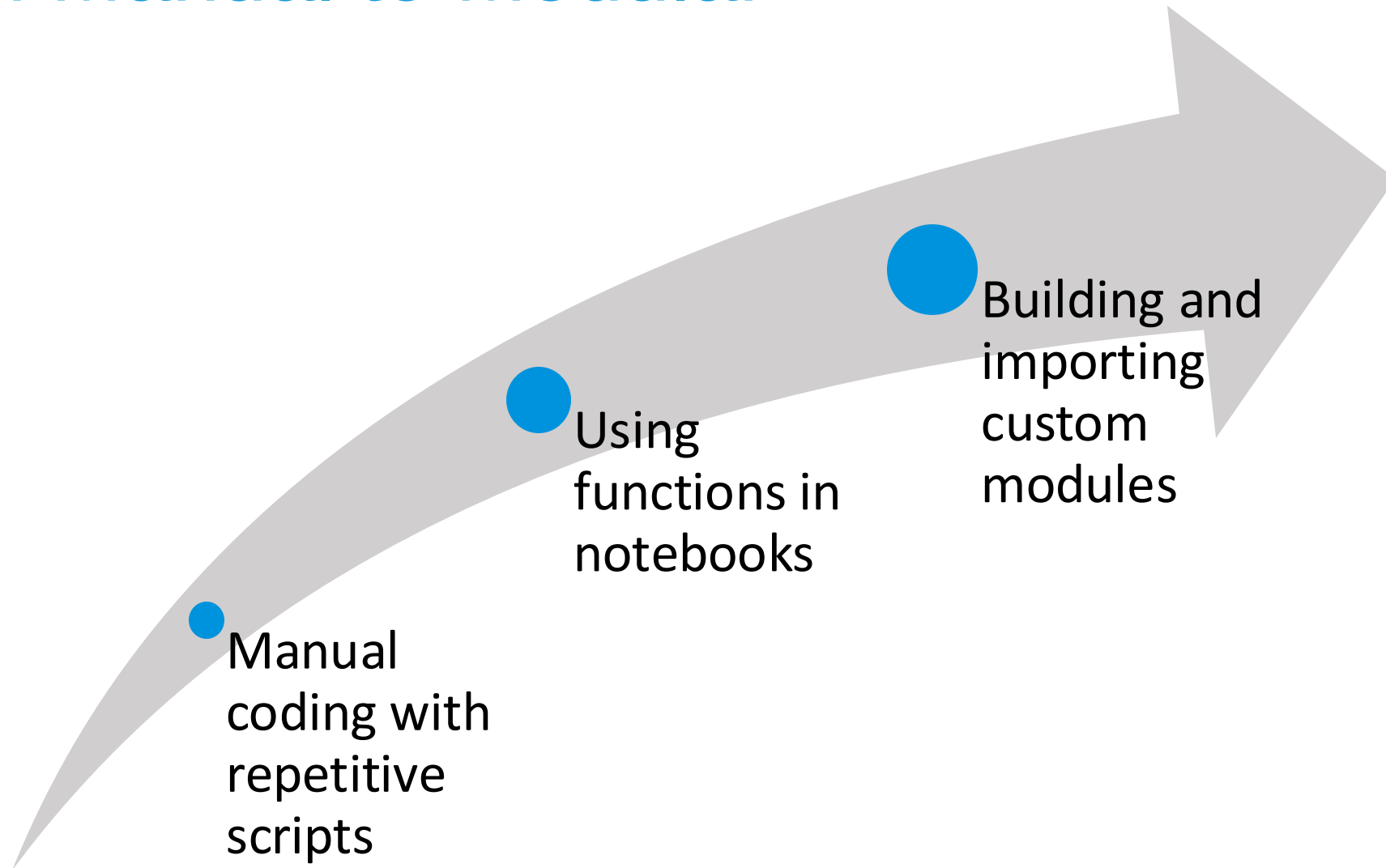
- Finding and using common functions and procedures
- Sharing modules across projects
- Benefits:
 - Saves time
 - Reduces errors
 - Simplifies updates

Why use custom modules?



- Reusability: write once, use everywhere
- Consistency: standardises processes across projects
- Efficiency: accelerates development cycles (less time spent on 'reinventing the wheel')
- Collaboration: easier for teams to work together

From Manual to Modular



Stage 1 – Manual Coding



```
# Import necessary PySpark functions
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, to_date

# Initialize SparkSession (Assuming this is done in Microsoft Fabric Lakehouse environment)
spark = SparkSession.builder.getOrCreate()

# Load data from CSV files for three different departments
df_sales = spark.read.csv("/lakehouse/data/sales_data.csv", header=True, inferSchema=True)
df_marketing = spark.read.csv("/lakehouse/data/marketing_data.csv", header=True, inferSchema=True)
df_hr = spark.read.csv("/lakehouse/data/hr_data.csv", header=True, inferSchema=True)

# Transform sales data
df_sales = df_sales.withColumn("date", to_date(col("date"), "MM/dd/yyyy"))
df_sales = df_sales.withColumn("revenue", col("revenue").cast("double"))
df_sales = df_sales.filter(col("revenue") > 0)
df_sales = df_sales.withColumnRenamed("revenue", "amount")
df_sales = df_sales.select("date", "amount", "product_id", "customer_id")

# Transform marketing data
df_marketing = df_marketing.withColumn("date", to_date(col("date"), "MM/dd/yyyy"))
df_marketing = df_marketing.withColumn("cost", col("cost").cast("double"))
df_marketing = df_marketing.filter(col("cost") > 0)
df_marketing = df_marketing.withColumnRenamed("cost", "amount")
df_marketing = df_marketing.select("date", "amount", "campaign_id")

# Transform HR data
df_hr = df_hr.withColumn("date", to_date(col("date"), "MM/dd/yyyy"))
df_hr = df_hr.withColumn("salary", col("salary").cast("double"))
df_hr = df_hr.filter(col("salary") > 0)
df_hr = df_hr.withColumnRenamed("salary", "amount")
df_hr = df_hr.select("date", "amount", "employee_id")
```

Stage 1 – Manual Coding



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# Transform HR data
df_hr = df_hr.withColumn("date", to_date(col("date"), "MM/dd/yyyy"))
df_hr = df_hr.withColumn("salary", col("salary").cast("double"))
df_hr = df_hr.filter(col("salary") > 0)
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df_marketing = df_marketing.withColumnRenamed("cost", "amount")
df_marketing = df_marketing.select("date", "amount", "campaign_id")

# Transform HR data
df_hr = df_hr.withColumn("date", to_date(col("date"), "MM/dd/yyyy"))
df_hr = df_hr.withColumn("salary", col("salary").cast("double"))
df_hr = df_hr.filter(col("salary") > 0)
df_hr = df_hr.withColumnRenamed("salary", "amount")
df_hr = df_hr.select("date", "amount", "employee_id")
```

Stage 2 – Using Functions



```
# Import necessary PySpark functions
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, to_date

# Initialize SparkSession (Assuming this is done in Microsoft Fabric Lakehouse environment)
spark = SparkSession.builder.getOrCreate()

# Define a function to read CSV files
def read_csv_file(file_path):
    return spark.read.csv(file_path, header=True, inferSchema=True)

# Define a function to transform DataFrames
def transform_dataframe(df, amount_column):
    df = df.withColumn("date", to_date(col("date"), "MM/dd/yyyy"))
    df = df.withColumn("amount", col(amount_column).cast("double"))
    df = df.filter(col("amount") > 0)
    df = df.select("date", "amount", *[col for col in df.columns if col not in ["date", amount_column]])
    return df

# List of datasets with their respective file paths and amount columns
datasets = [
    {"name": "sales", "file_path": "/lakehouse/data/sales_data.csv", "amount_column": "revenue"},
    {"name": "marketing", "file_path": "/lakehouse/data/marketing_data.csv", "amount_column": "cost"},
    {"name": "hr", "file_path": "/lakehouse/data/hr_data.csv", "amount_column": "salary"},
]

# Read and transform each dataset using the defined functions
transformed_dfs = []

for dataset in datasets:
    df = read_csv_file(dataset["file_path"])
    df = transform_dataframe(df, dataset["amount_column"])
    transformed_dfs.append(df)
```

Stage 2 – Using Functions



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    df = df.withColumn("amount", col(amount_column).cast("double"))
    df = df.filter(col("amount") > 0)
    df = df.select("date", "amount", *[col for col in df.columns if col not in ["date", amount_column]])
    return df

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    {"name": "marketing", "file_path": "/lakehouse/data/marketing_data.csv", "amount_column": "cost"},
    {"name": "hr", "file_path": "/lakehouse/data/hr_data.csv", "amount_column": "salary"},
]

# Read and transform each dataset using the defined functions
transformed_dfs = []

for dataset in datasets:
    df = read_csv_file(dataset["file_path"])
    df = transform_dataframe(df, dataset["amount_column"])
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Stage 2 – Using Functions



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# Import necessary PySpark functions
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, to_date

# Initialize SparkSession (Assuming this is done in Microsoft Fabric Lakehouse environment)
spark = SparkSession.builder.getOrCreate()

# Define a function to read CSV files
def read_csv_file(file_path):
    return spark.read.csv(file_path, header=True, inferSchema=True)

# Define a function to transform DataFrames
def transform_dataframe(df, amount_column):
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    return df

# List of datasets with their respective file paths and amount columns
datasets = [
    {"name": "sales", "file_path": "/lakehouse/data/sales_data.csv", "amount_column": "revenue"},
    {"name": "marketing", "file_path": "/lakehouse/data/marketing_data.csv", "amount_column": "cost"},
    {"name": "hr", "file_path": "/lakehouse/data/hr_data.csv", "amount_column": "salary"},
]

# Read and transform each dataset using the defined functions
transformed_dfs = []

for dataset in datasets:
    df = read_csv_file(dataset["file_path"])
    df = transform_dataframe(df, dataset["amount_column"])
    transformed_dfs.append(df)
```

Stage 3 – Using Custom Modules



```
# Import necessary PySpark functions
from pyspark.sql.functions import col, to_date

def read_csv_file(spark, file_path):
    """
    Reads a CSV file into a Spark DataFrame.

    Parameters:
    - spark: The SparkSession object.
    - file_path (str): The path to the CSV file.

    Returns:
    - DataFrame: The loaded DataFrame.
    """
    return spark.read.csv(file_path, header=True, inferSchema=True)

def transform_dataframe(df, amount_column):
    """
    Transforms a DataFrame by processing date and amount columns.

    Parameters:
    - df (DataFrame): The DataFrame to transform.
    - amount_column (str): The name of the column representing the amount.

    Returns:
    - DataFrame: The transformed DataFrame.
    """
    df = df.withColumn("date", to_date(col("date"), "MM/dd/yyyy"))
    df = df.withColumn("amount", col(amount_column).cast("double"))
    df = df.filter(col("amount") > 0)
    df = df.select("date", "amount", *[col for col in df.columns if col not in ["date", amount_column]])
    return df
```

Stage 3 – Using Custom Modules



```
# Import necessary PySpark functions and custom module
from pyspark.sql import SparkSession
from data_utils import read_csv_file, transform_dataframe

# Initialize SparkSession (Assuming this is done in Microsoft Fabric Lakehouse environment)
spark = SparkSession.builder.getOrCreate()

# List of datasets with their respective file paths and amount columns
datasets = [
    {"name": "sales", "file_path": "/lakehouse/data/sales_data.csv", "amount_column": "revenue"},
    {"name": "marketing", "file_path": "/lakehouse/data/marketing_data.csv", "amount_column": "cost"},
    {"name": "hr", "file_path": "/lakehouse/data/hr_data.csv", "amount_column": "salary"},
]

# Read and transform each dataset using the imported functions
transformed_dfs = []

for dataset in datasets:
    df = read_csv_file(spark, dataset["file_path"])
    df = transform_dataframe(df, dataset["amount_column"])
    transformed_dfs.append(df)
```

Stage 3 – Using Custom Modules



```
# Import necessary PySpark functions
from pyspark.sql.functions import col, to_date

def read_csv_file(spark, file_path):
    """
    Reads a CSV file into a Spark DataFrame.

    Parameters:
    - spark: The SparkSession object.
    - file_path (str): The path to the CSV file.

    Returns:
    - DataFrame: The loaded DataFrame.
    """
    return spark.read.csv(file_path, header=True, inferSchema=True)

def transform_dataframe(df, amount_column):
    """
    Transforms a DataFrame by processing date and amount columns.

    Parameters:
    - df (DataFrame): The DataFrame to transform.
    - amount_column (str): The name of the column representing the amount.

    Returns:
    - DataFrame: The transformed DataFrame.
    """
    df = df.withColumn("date", to_date(col("date"), "MM/dd/yyyy"))
    df = df.withColumn("amount", col(amount_column).cast("double"))
    df = df.filter(col("amount") > 0)
    df = df.select("date", "amount", *[col for col in df.columns if col not in ("date", "amount")])
    return df
```

```
# Import necessary PySpark functions and custom module
from pyspark.sql import SparkSession
from data_utils import read_csv_file, transform_dataframe

# Initialize SparkSession (Assuming this is done in Microsoft Fabric Lakehouse environment)
spark = SparkSession.builder.getOrCreate()

# List of datasets with their respective file paths and amount columns
datasets = [
    {"name": "sales", "file_path": "/lakehouse/data/sales_data.csv", "amount_column": "revenue"},
    {"name": "marketing", "file_path": "/lakehouse/data/marketing_data.csv", "amount_column": "cost"},
    {"name": "hr", "file_path": "/lakehouse/data/hr_data.csv", "amount_column": "salary"},
]

# Read and transform each dataset using the imported functions
transformed_dfs = []

for dataset in datasets:
    df = read_csv_file(spark, dataset["file_path"])
    df = transform_dataframe(df, dataset["amount_column"])
    transformed_dfs.append(df)
```



Live demo

Process



Python project using VS Code



Package to .whl file



Publish to Fabric environment



Import in notebook



Call and execute methods

Demo time!





What we saw

- Python project & wheel file creation
- Importing into Fabric
- Calling the methods we created



In summary

- Reusable code
 - Write once, use many times
- Less lines of code
 - Reduces technical debt
 - Easier maintenance
- Easier to force development guidelines
 - No 'handwriting' for a single developer

Key takeaways



- DRY principle enhances efficiency in software development
- We can take this into data engineering
- Custom Python modules are powerful tools in Microsoft Fabric
- Implementing an abstract framework makes it easier to develop and maintain your Fabric solution

Bas Land



- BI consultant since 2013
- Co-founder of Kimura Data Intelligence
 - Standardised framework for Fabric
 - Analytical products for several software partners
 - Consultancy (Fabric & Power BI)
- Married to Anouk, we have a dachshund (😊) called Chester, and a baby on the way
- Sports: purple belt Brazilian jiu-jitsu, weight lifting, running



Contact



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ThatFabricGuy



LinkedIn



Questions?